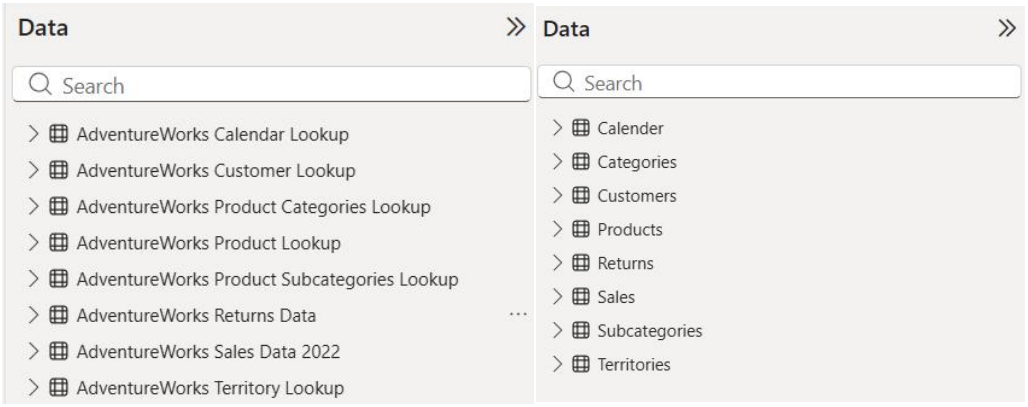
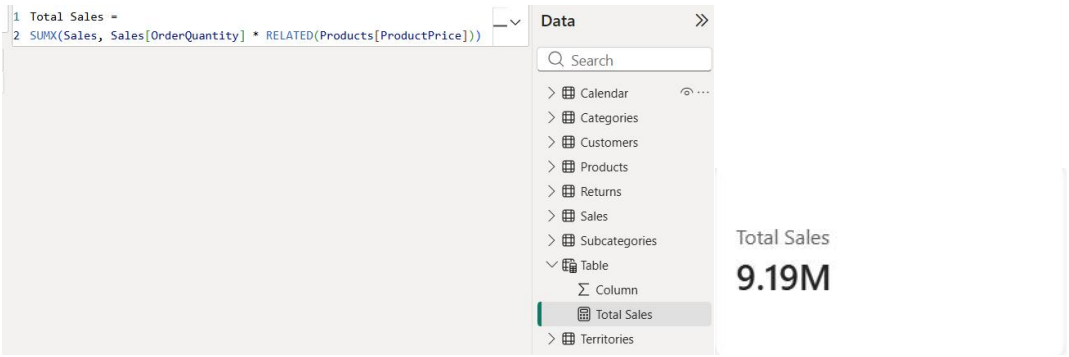
Data Visualization Lab

Assignment 4

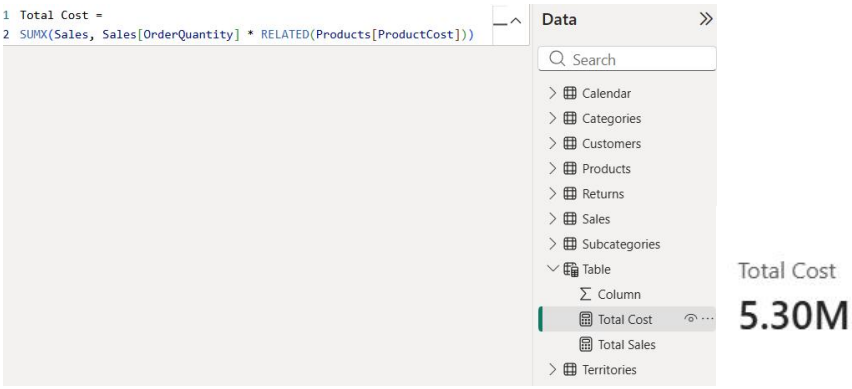
DAX Calculations



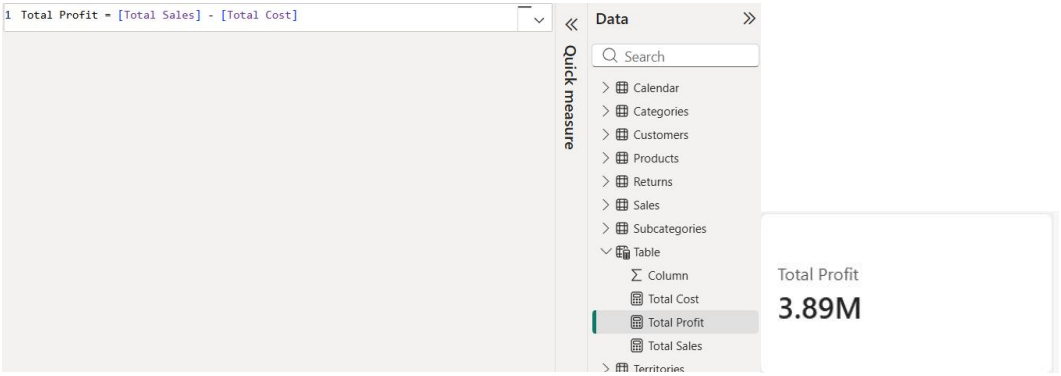
Load necessary tables and rename convenience



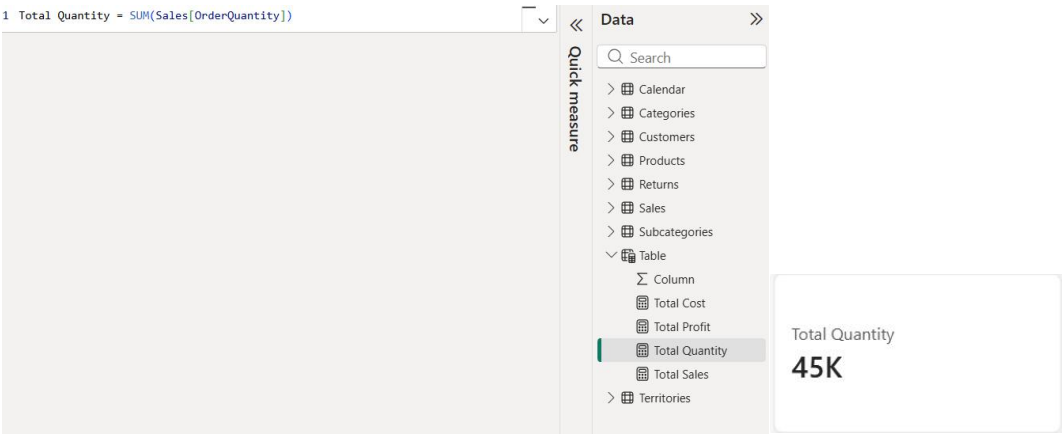
Multiplies quantity by product price for every order line, then adds them all up.



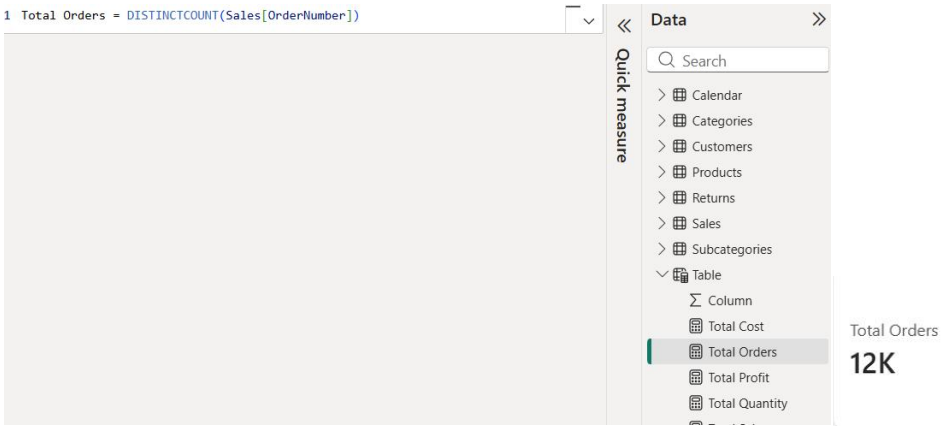
Multiplies quantity by product cost for every order line, then adds them up.



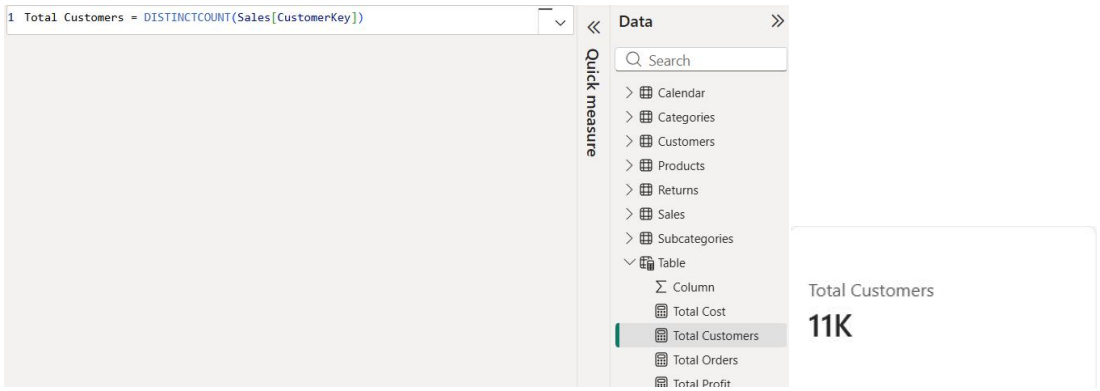
Subtracts Total Cost from Total Sales.



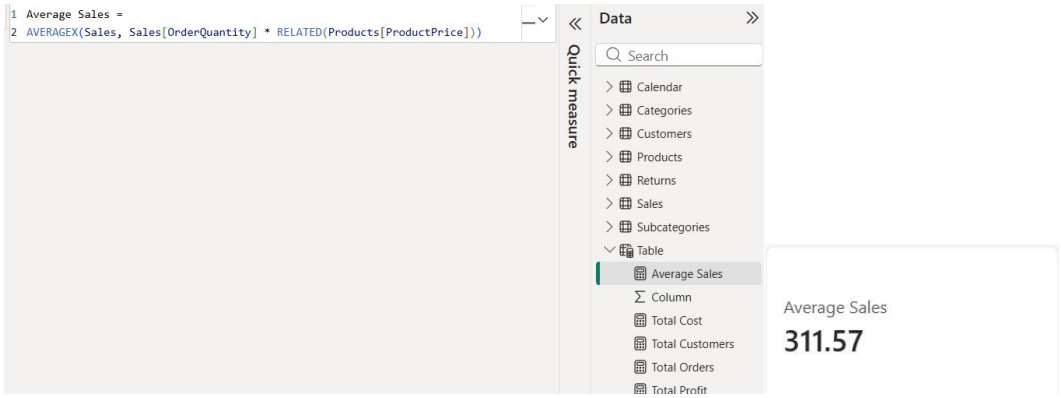
Adds up all items ordered.



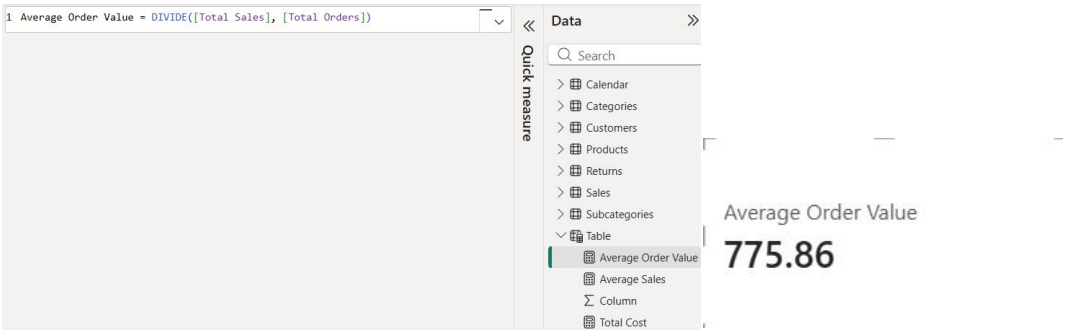
Counts each unique order number once.



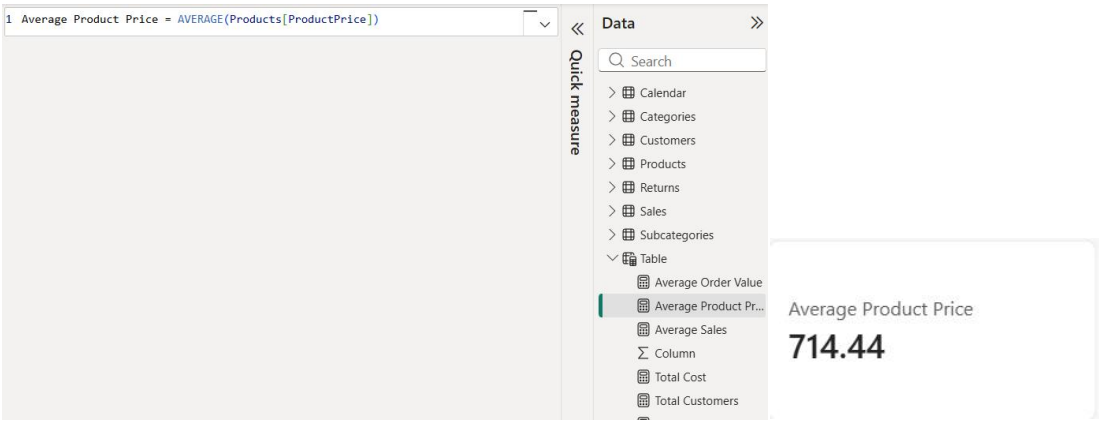
Counts each unique customer once.



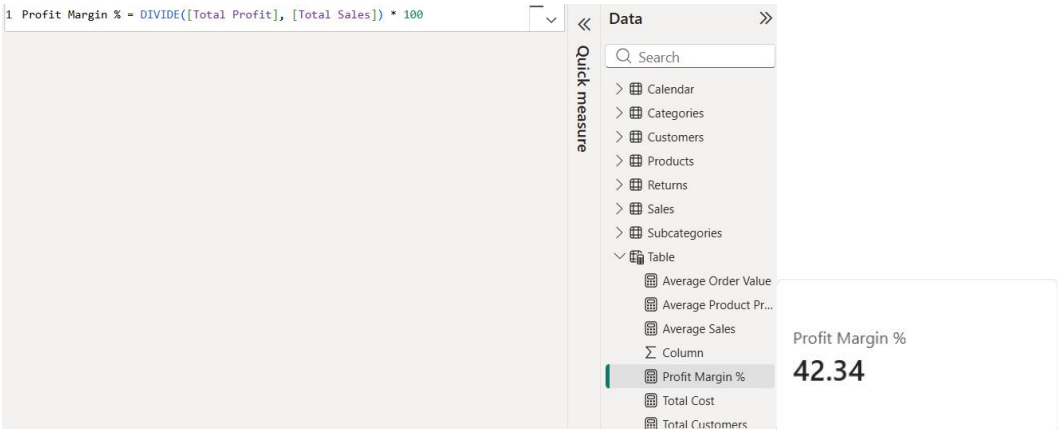
Gives the average sales value of one order line.



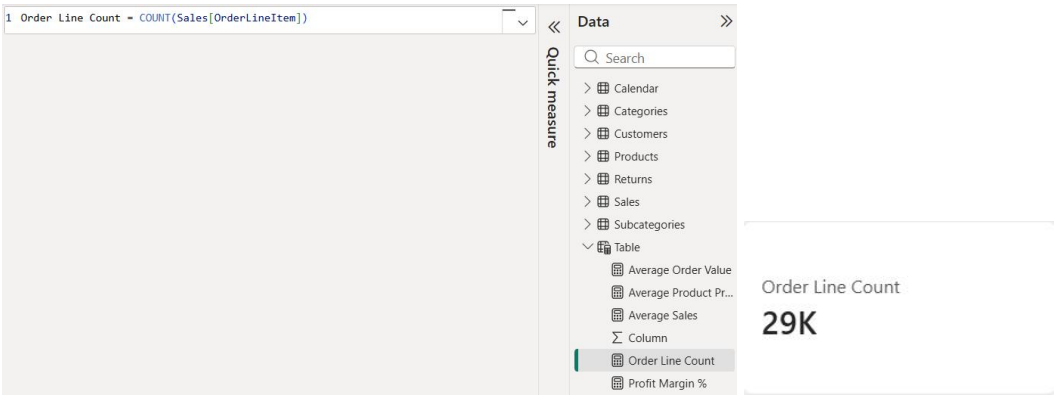
Divides Total Sales by Total Orders to get the average money per order.



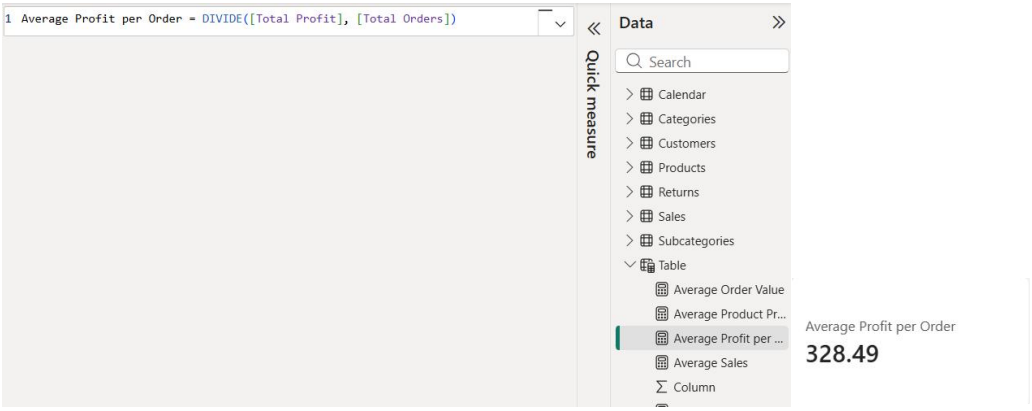
Gives the average selling price across all products.



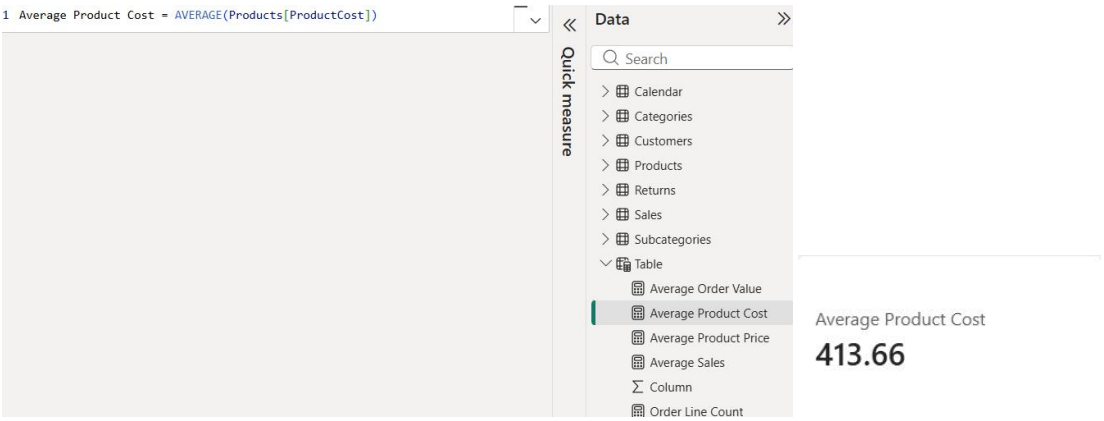
Divides profit by sales and multiplies by 100 to show how much of each sale is profit.



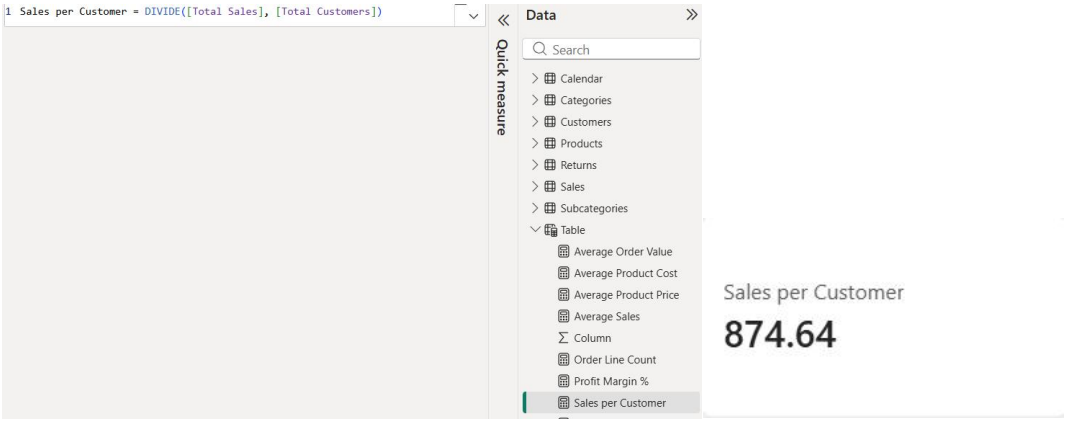
Counts the number of order line items.



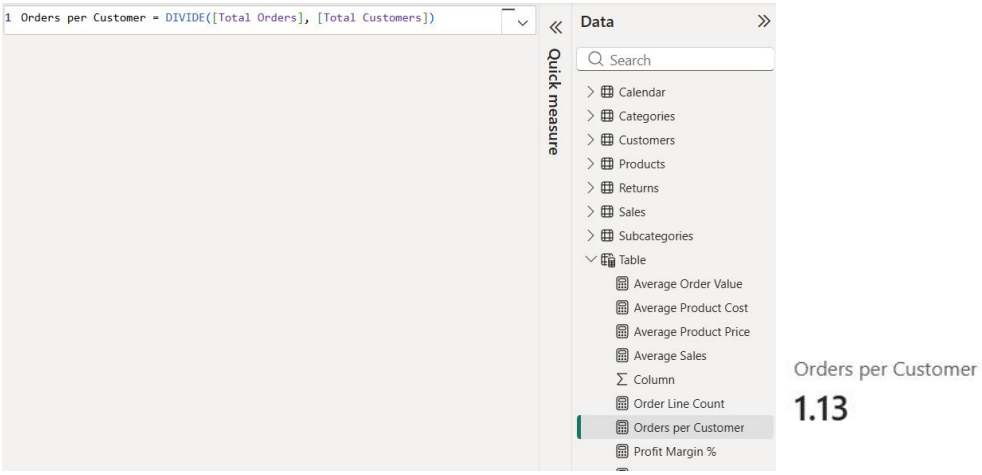
Divides Total Profit by Total Orders.



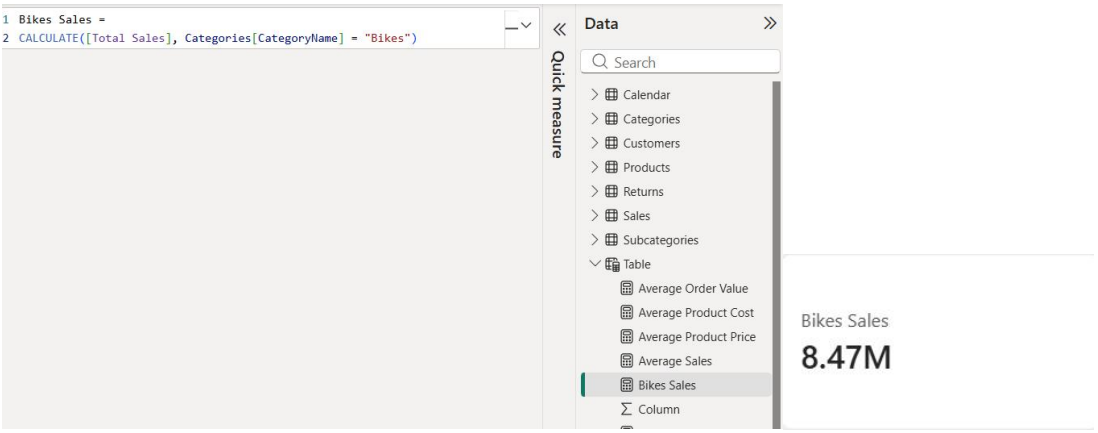
Gives the average cost price of products.



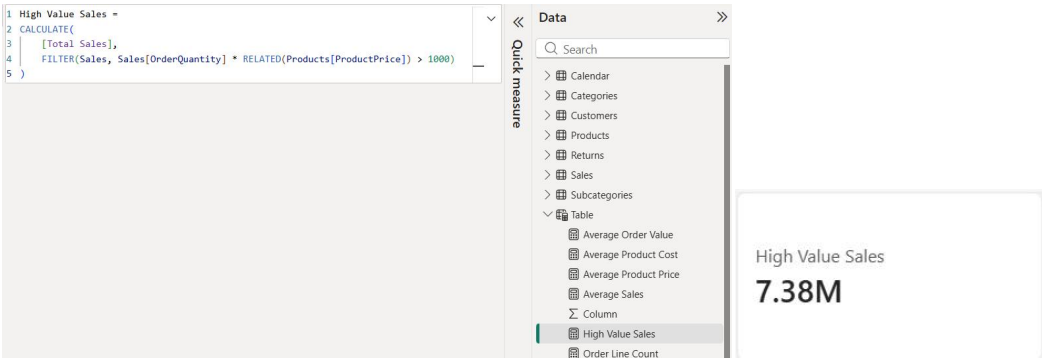
Divides Total Sales by Total Customers.



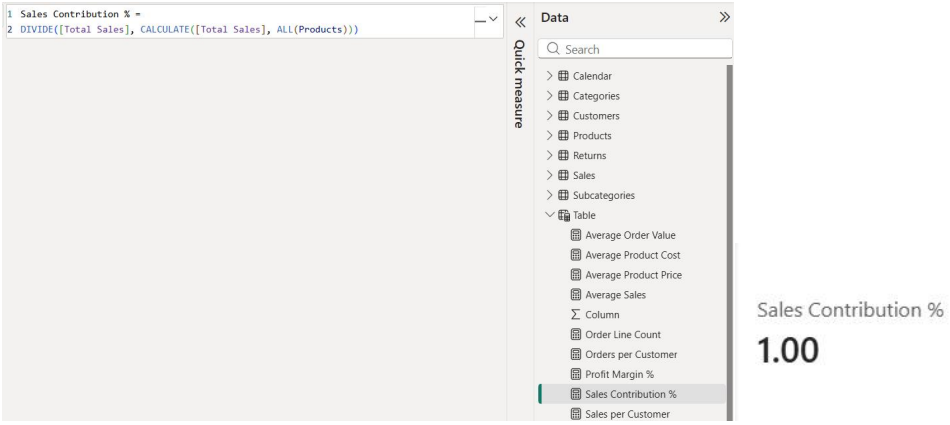
Shows how many orders each customer places on average.



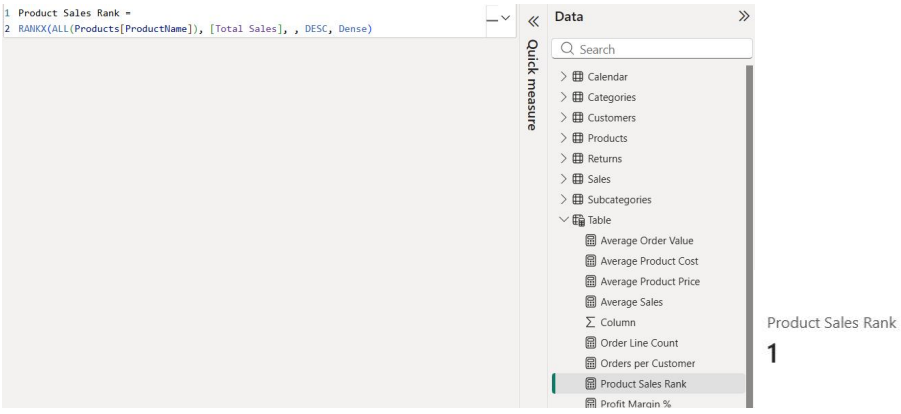
Shows sales for the Bikes category only.



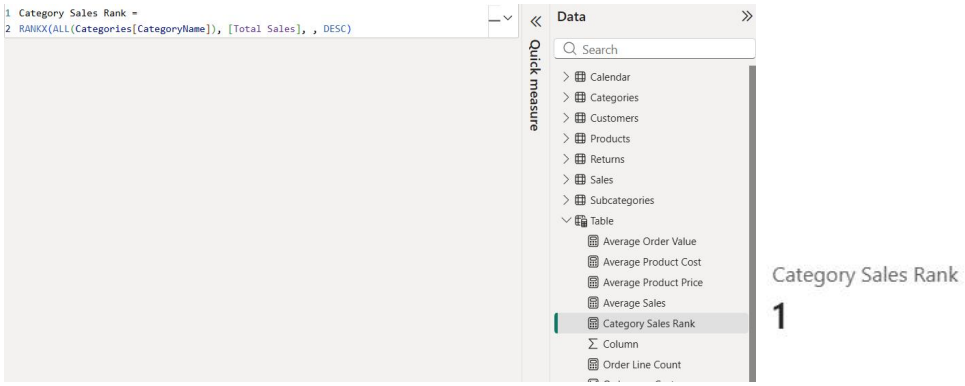
Adds up only the order lines worth more than 1000.



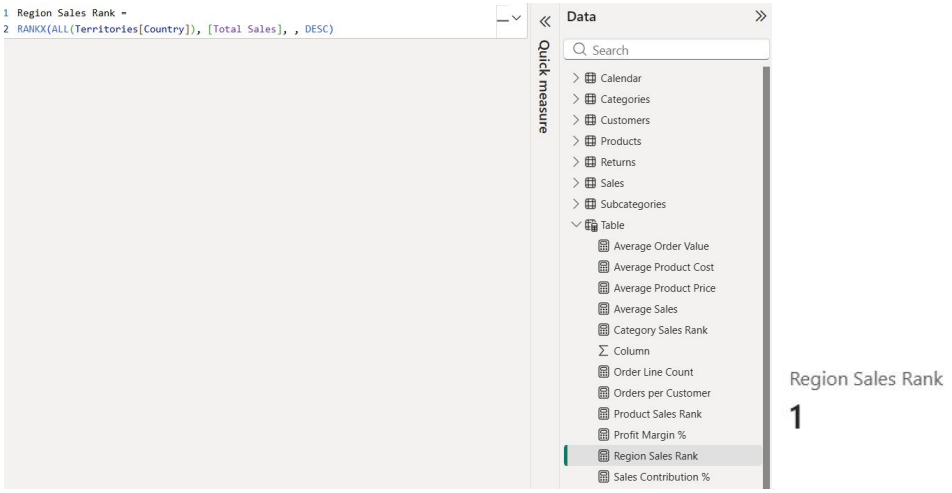
Shows each item’s share of total sales.



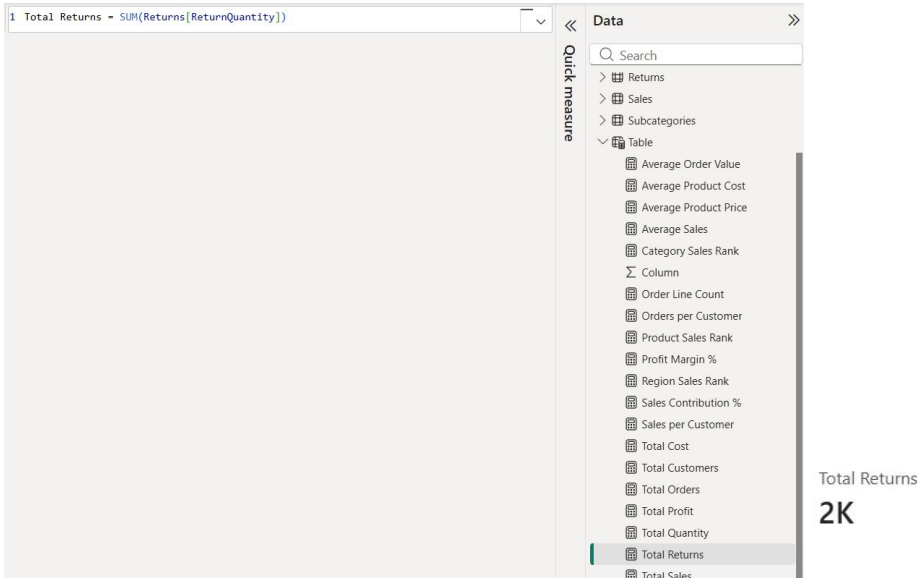
Ranks products from highest to lowest sales.



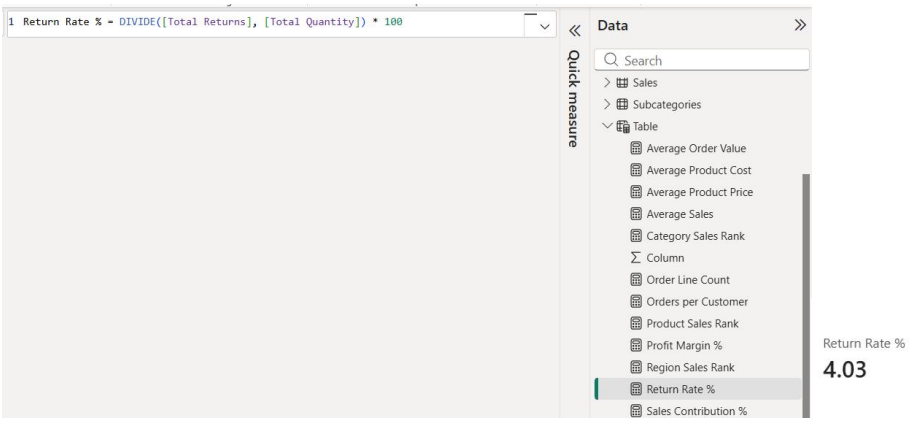
Ranks categories from highest to lowest sales.



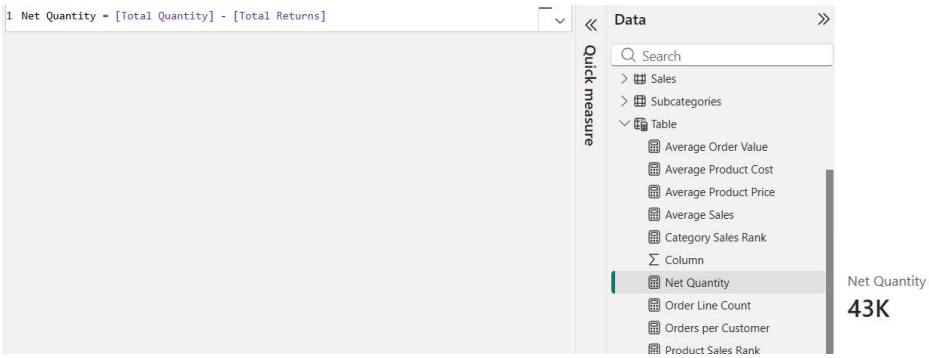
Ranks countries from highest to lowest sales.



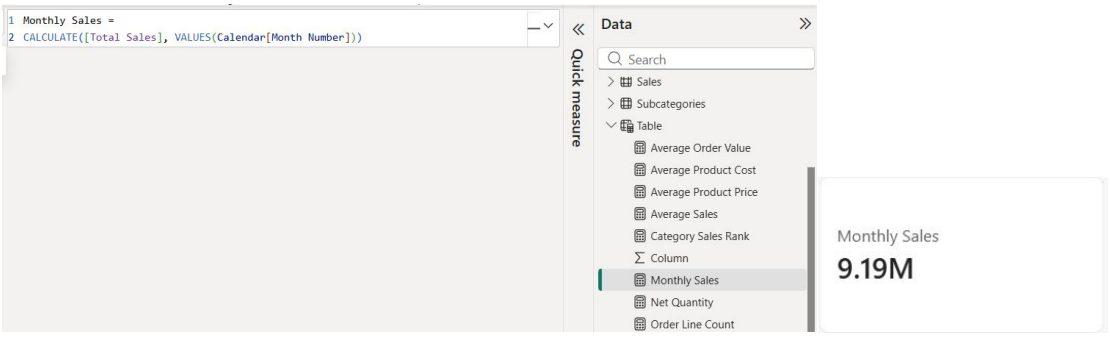
Adds up all returned quantities.



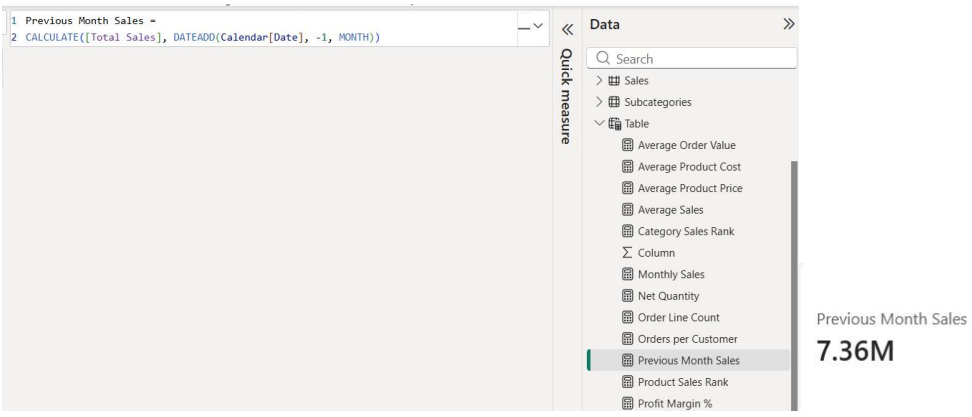
Shows what percentage of sold items were returned.



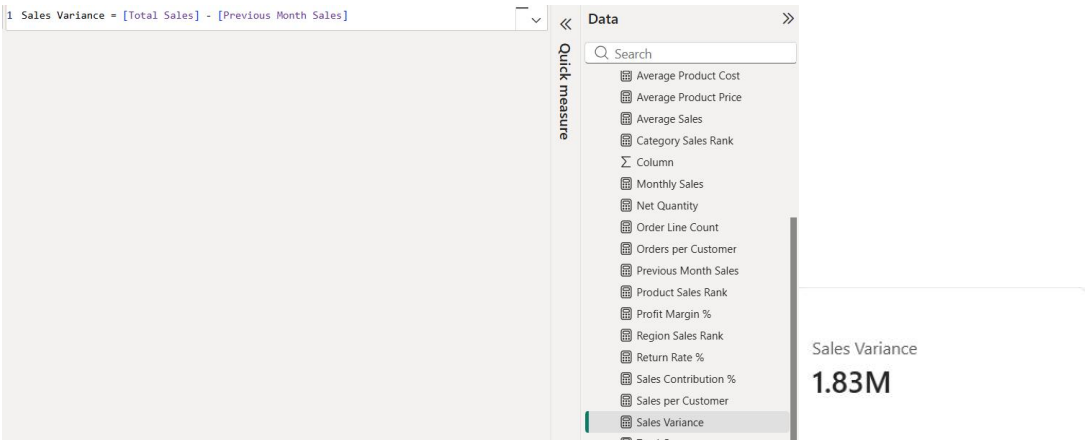
Subtracts returns from the quantity sold.



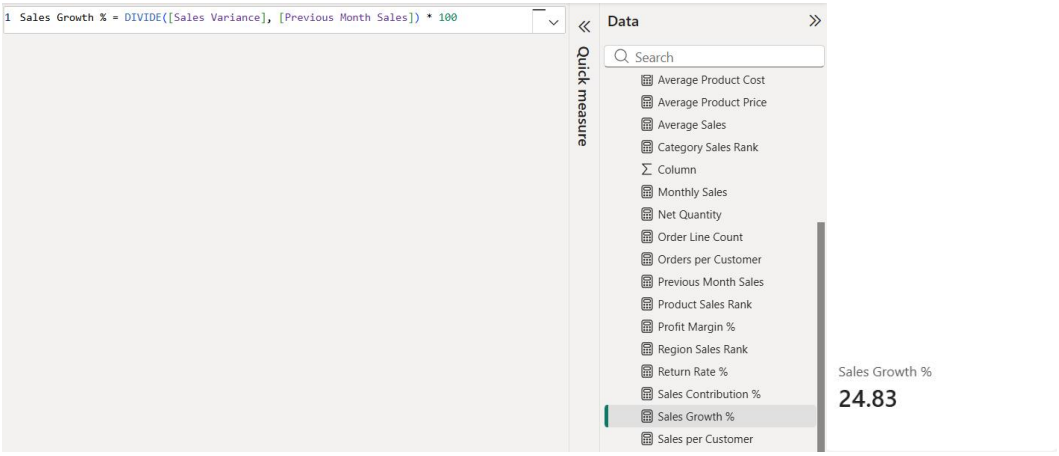
Shows total sales for each month.



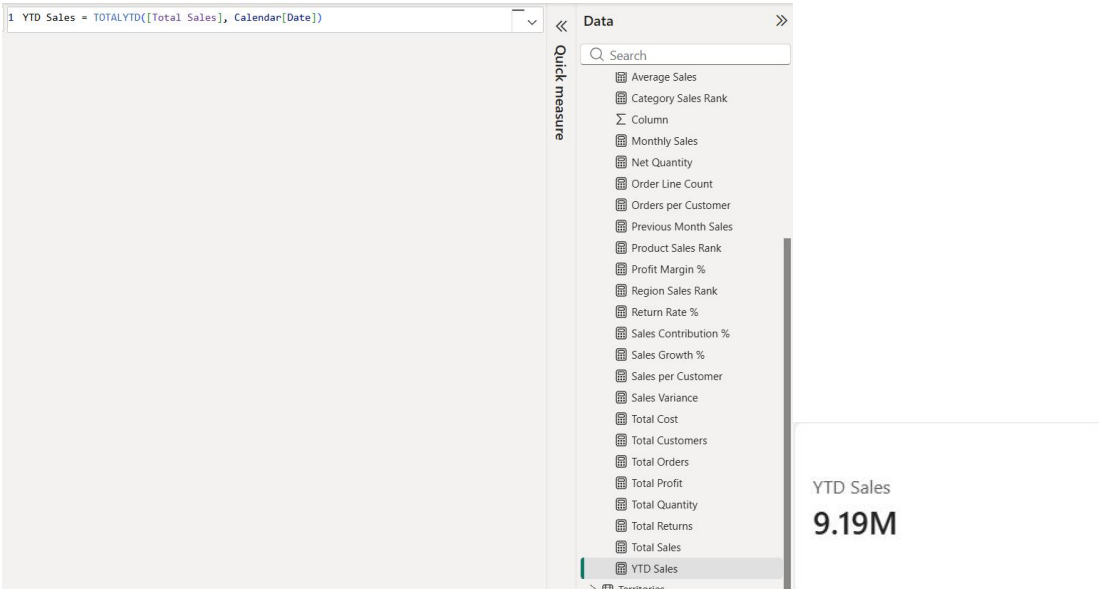
Shows the sales of the month before the current one.



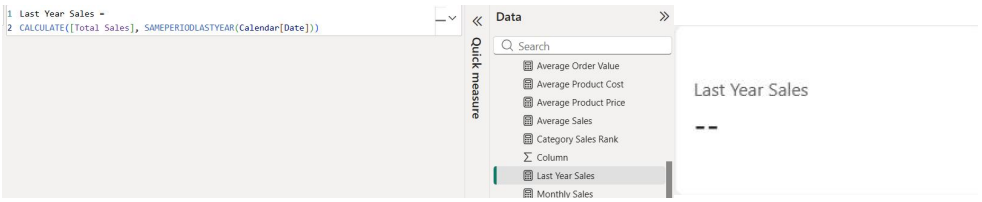
Shows the difference between this month and last month.



Shows that difference as a percentage of last month.



Adds up sales from January to the current date in the year.



Shows sales for the same period last year.Here,the sales table is only for 2022 year

Dax Queries :

```
-- Q1: KPI validation
EVALUATE
ROW(
    "Total Sales", [Total Sales],
    "Total Cost", [Total Cost],
    "Total Profit", [Total Profit],
    "Profit Margin %", [Profit Margin %],
    "Total Orders", [Total Orders],
    "Total Customers", [Total Customers]
)
```

Results | Result 1 of 1 | Copy

	[Total Sales]	[Total Cost]	[Total Profit]	[Profit Margin %]	[Total Orders]	[Total Customers]
1	9185449.45	5296420.47	3889028.97	42.34	11839	10502

```
-- Q2: sales and profit by category
EVALUATE
SUMMARIZECOLUMNS(
    Categories[CategoryName],
    "Sales", [Total Sales],
    "Profit", [Total Profit],
    "Margin %", [Profit Margin %]
)
ORDER BY [Sales] DESC
```

Results | Result 1 of 1 | Copy

	Categories[Category...	[Sales]	[Profit]	[Margin %]
1	Bikes	8468854.53	3477544.92	41.06
2	Accessories	507330.99	318823.94	62.84
3	Clothing	209263.93	92660.11	44.28

```dax
-- Q3: top 10 products
EVALUATE
TOPN(
    10,
    SUMMARIZECOLUMNS(Products[ProductName], "Sales", [Total Sales]),
    [Sales], DESC
)
```

Results Result 1 of 1 Copy

	Products[ProductNa...	[Sales]
1	Road-350-W Yellow, 42	258550.48
2	Road-350-W Yellow, 40	261952.46
3	Touring-1000 Yellow, 46	264631.77
4	Touring-1000 Blue, 46	281320.26
5	Mountain-200 Black, 46	465145.29
6	Mountain-200 Black, 42	504078.16
7	Mountain-200 Black, 38	514323.65
8	Mountain-200 Silver, 46	505426.38
9	Mountain-200 Silver, 38	513712.06
10	Mountain-200 Silver, 42	466069.41

```dax
-- Q4: monthly trend with previous period
EVALUATE
SUMMARIZECOLUMNS(
    Calendar[Year], Calendar[Month Number], Calendar[Month Name],
    "Sales", [Total Sales],
    "Prev Month", [Previous Month Sales],
    "Growth %", [Sales Growth %]
)
ORDER BY Calendar[Year], Calendar[Month Number]
```

Results Result 1 of 1 Copy

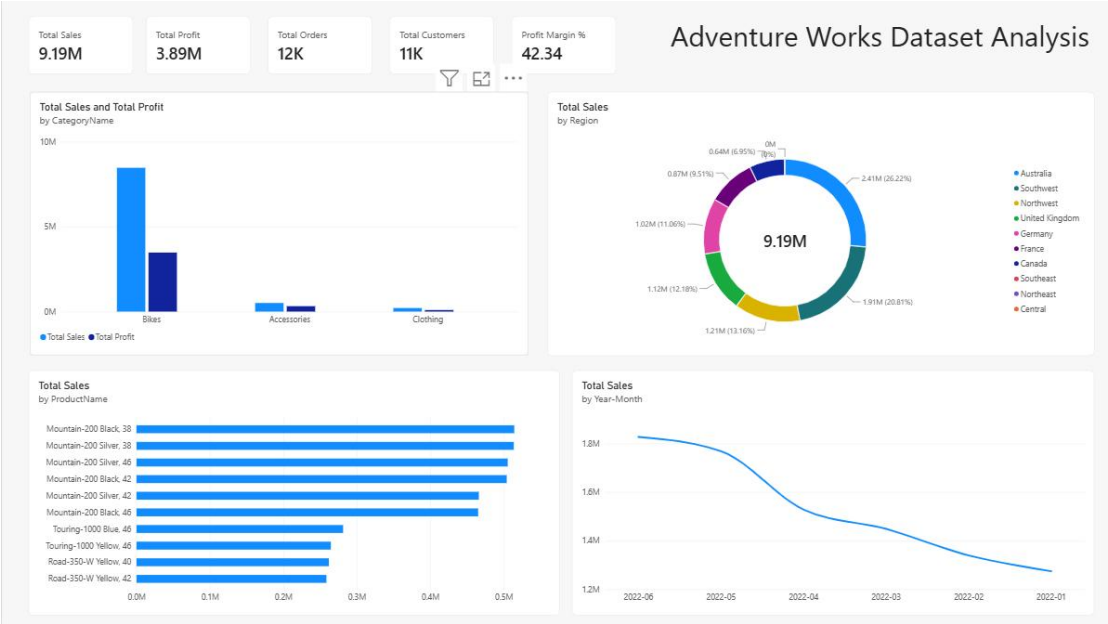
	Calendar[Year]	Calendar[Month Nu...	Calendar[Month Na...	[Sales]	[Prev Month]	[Growth %]
1	2022	1	Jan	1274378.67		
2	2022	2	Feb	1339241.29	1274378.67	5.09
3	2022	3	Mar	1448596.12	1339241.29	8.17
4	2022	4	Apr	1527813.72	1448596.12	5.47
5	2022	5	May	1768432.51	1527813.72	15.75
6	2022	6	Jun	1826987.14	1768432.51	3.31

```dax
-- Q5: sales by region
EVALUATE
SUMMARIZECOLUMNS(
    Territories[Region], Territories[Country],
    "Sales", [Total Sales],
    "Profit", [Total Profit]
)
ORDER BY [Sales] DESC
```

Results Result 1 of 1 Copy

	Territories[Region]	Territories[Country]	[Sales]	[Profit]
1	Australia	Australia	2408104.87	995203.01
2	Southwest	United States	1911216.91	815381.98
3	Northwest	United States	1208890.84	519154.09
4	United Kingdom	United Kingdom	1119087.76	472552.24
5	Germany	Germany	1015657.33	430885.3
6	France	France	873754.28	370048.69
7	Canada	Canada	638086.09	280976.76
8	Southeast	United States	7941.04	3499.2
9	Northeast	United States	2402.57	1157.7
10	Central	United States	307.74	170.01

Dashboard:


Total Sales
9.19M
Total Profit
3.89M
Total Orders
12K
Total Customers
11K
Profit Margin %
42.34
Adventure Works Dataset Analysis
Total Sales and Total Profit
by CategoryName
10M
5M
0M
Bikes
Accessories
Clothing
Total Sales
Total Profit
Total Sales
by Region
9.19M
2.41M (26.22%)
1.91M (20.81%)
1.21M (13.16%)
1.12M (12.18%)
1.02M (11.06%)
0.87M (9.51%)
0.64M (6.95%)
0M (0%)
Australia
Southwest
Northwest
United Kingdom
Germany
France
Canada
Southeast
Northeast
Central
Total Sales
by ProductName
Mountain-200 Black, 38
Mountain-200 Silver, 38
Mountain-200 Silver, 46
Mountain-200 Black, 42
Mountain-200 Silver, 42
Mountain-200 Black, 46
Touring-1000 Blue, 46
Touring-1000 Yellow, 46
Road-350-W Yellow, 40
Road-350-W Yellow, 42
0.0M
0.1M
0.2M
0.3M
0.4M
0.5M
Total Sales
by Year-Month
1.8M
1.6M
1.4M
1.2M
2022-06
2022-05
2022-04
2022-03
2022-02
2022-01